

Organozinc alkoxide-promoted aldol-Tishchenko reaction of aliphatic aldehydes: an expedient entry to prepare the α -methylene ketones

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Abstract—*i*-PrOZnEt is an excellent reagent to promote the aldol-Tishchenko reaction of the aliphatic aldehydes tethered with other labile functional groups. The 1,3-diol monoesters **4** were formed as the major products, which could be converted to α -methylene ketones **7** in two steps in good yields.

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1. Introduction

For enolizable aldehydes, both aldol addition and Tishchenko reaction are two major possible pathways. For the aldol addition, the presence of only basic sites in the catalysts is sufficient, whereas for the Tishchenko reaction, the presence of both acidic sites and basic sites in the catalysts is required. Interestingly, sequential aldol-Tishchenko reaction can become competing if the catalyst first, accomplishes the aldol reaction, which is followed by Tishchenko esterification by the Lewis acidic nature of the same catalyst. In many cases, traditional Tishchenko¹ and aldol-Tishchenko reactions² are competing with each other.³ However, sequential aldol-Tishchenko reaction can become competing if the catalyst is sufficiently basic, such as a basic metal hydroxide catalyst.^{4,5} Some other catalysts, such as metal alkoxides of monofunctional alcohols,^{3a,6} simple metal hydroxides.⁶ Lithium monoalcoholate of 1,3-diol,⁷ LiWO₂,⁸ Cp*2Sm(thf)₂,⁹ polynuclear carbonyl ferrate,¹⁰ and arylmagnesium halide in HMPT¹¹ have also been used. The aldol-Tishchenko reaction of the aldehyde gives 1,3-diol monoesters from the trimerization of the aldehyde. Trimerization of isobutyraldehyde to 2,2,4-trimethylpentane-1,3-diol mono-isobutyrate, which are the most common coalescing agents, for example, in latex paints,¹² has been industrialized since, 1988 by use of binary metal oxide BaO–CaO.¹³

Keywords: Organozinc alkoxide; Aldol-Tishchenko reaction; Tishchenko reaction; 1,3-Diol monoester; α -Methylene ketones.

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We found that the aliphatic aldehyde **1a** reacted with a catalytic amount of diisobutylaluminium hydride (Dibal-H) in pentane to give ester **2a** in good to excellent yields via a Tishchenko reaction intermediate **A** in which an intramolecular hydride shift took place. The aluminium alkoxide **B** formed in the reaction reacted with aldehyde **1a** to establish the catalytic reaction cycle (Fig. 1).¹⁴ This observation provokes us to investigate the influence of the metal ions on the metal alkoxides-promoted Tishchenko reactions. Interestingly, we found that the change of the metal ion of the alkoxide **B** will switch the reaction pathway from Tishchenko reaction to aldol-Tishchenko reaction. Herein, we wish to report the results of our studies on the homogeneous aldol-Tishchenko reaction promoted by magnesium and zinc alkoxides. The further synthetic

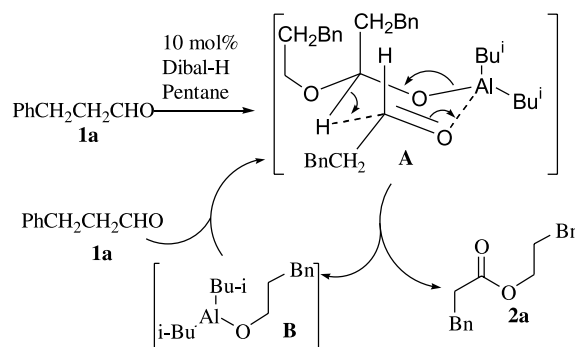


Figure 1. The plausible mechanism for the reaction of aldehyde **1a** with catalytic amount of Dibal-H to give the corresponding ester **2a**.

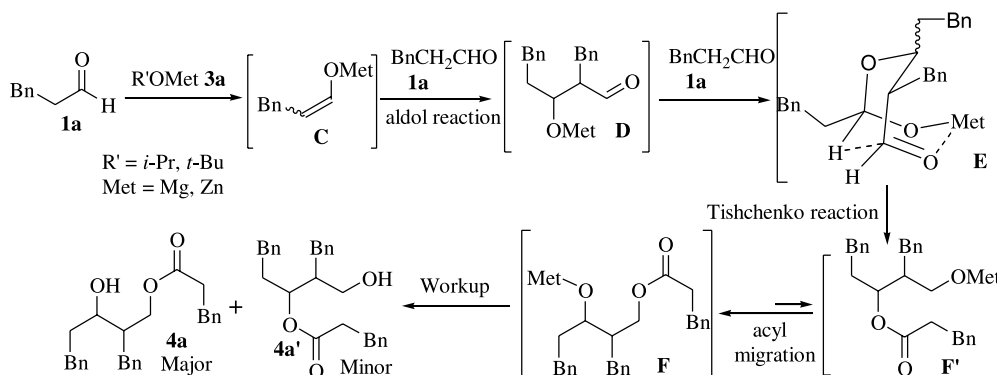


Figure 2. The proposed aldol-Tishchenko reaction mechanism for the formation of compounds **4a** and **4a'** from 3-phenylpropanal (**1a**) promoted by metal alkoxides.

applications of their products in the synthesis of the α -methylene ketones will also be discussed.

2. Results and discussions

2.1. The reaction pathway is metal ion-dependent in the reactions of aliphatic aldehydes with metal alkoxides

A solution of 0.1 mol equiv of $\text{Ph}(\text{CH}_2)_3\text{OMgCl}$ (**3a**), prepared from equal mol of the 3-phenylpropyl alcohol with MeMgCl in THF at 0°C , was added dropwise in 5 min to 3-phenylpropanal (**1a**) in THF at 0°C . After the addition, the reaction was stirred at ambient temperature for 12 h. We found that no Tishchenko reaction product **2a** was formed and a mixture of 1,3-diol monoesters **4a'** and **4a** were isolated as major products (Fig. 2). The formation of these two compounds was rationalized as follows. An aldol reaction of the aldehyde first takes place and the aldol reaction product **D** forms a hemiacetal-like intermediate **E** with the free aldehyde. The hemiacetal-like intermediate **E** reacts to give the 1,3-diol monoester **4a'** via a Tishchenko reaction in which an intramolecular hydride shift takes place. The monoester intermediate **F'** is equilibrated with **F** via an acyl group migration (Fig. 2). Each of compounds **4a'** and **4a** contain two chiral centers. All these possible isomers are too close in polarity to be separated by silica gel column chromatography. In addition, it is very difficult to determine their ratio and identify, which is the major isomer from the 400 MHz ^1H NMR spectrum of the mixtures.

In order to find an optimal condition for the aldol-Tishchenko reaction, we tried to employ different kinds of metal alkoxides. We chose *i*-propanol or *tert*-butanol as reagents to generate the corresponding alkoxide based on their basicity, steric factor and readily availability. The

metal alkoxides were generated from the reaction of alcohols with phenylmagnesium chloride (2.0 M solution in THF) or diethylzinc (1.0 M solution in hexanes) at 0°C in THF. A solution of the metal alkoxide in THF was added dropwise in 5 min to 3-phenylpropanal (**1a**) in THF at 0°C . After the addition, the reaction was stirred at ambient temperature. A mixture of the 1,3-diol monoesters **4a'** and **4a** were obtained and their yield was shown in Table 1.

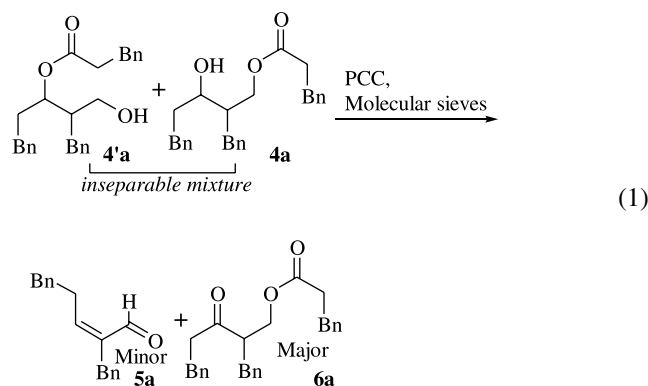
When *i*-PrOMgCl (0.33 mol equiv) was used as the promoter to react with aldehyde **1a** at 0°C , a mixture of the aldol-Tishchenko products **4a** and **4a'** were isolated in 81% yield (entry 1, Table 1). When the mol equiv of *i*-PrOMgCl was dropped to 0.17, the chemical yield of the aldol-Tishchenko products was slightly dropped to 75% (entry 2). When *t*-BuOMgCl (0.33 mol equiv) was used as the promoter, we isolated the aldol-Tishchenko products **4a** and **4a'** in 73% yield (entry 3). When the mol equiv of *t*-BuOMgCl was dropped to 0.17, the chemical yield of the aldol-Tishchenko products was decreased slightly to 61% (entry 4). Although the chemical yields are not improved by using *t*-BuOMgCl, their reaction times are shorter. When *i*-PrOZnEt (0.33 mol equiv) was used as the promoter, we isolated the aldol-Tishchenko products in 91% yield (entry 5). When the mol equiv of *i*-PrOZnEt was dropped to 0.17, the chemical yield of the aldol-Tishchenko products was decreased slightly to 73% (entry 6). The results of Table 1 indicate that *i*-PrOZnEt is an excellent reagent to promote the aldol-Tishchenko reaction from aliphatic aldehydes. The yield is high and the potential side reactions are minimized when 0.33 mol equiv of this reagent was used (entry 5). This conclusion is rationalized as follows. The Lewis acidity of the metal ion is responsible to the formation of the intermediate **E** in Figure 2. As for the Lewis acidity, zinc ion is stronger than that of magnesium ion. The zinc ion may be the better promoter to generate the hemiacetal intermediate **E** ($\text{Met}=\text{Zn}$) from the β -alkoxyaldehyde **D** and aldehyde **1a**. According to the literature report, the trimerization of aliphatic aldehydes to 1,3-diol monoesters is also effected by some sorts of metal alkoxides and phenoxides such as $\text{Ca}(\text{OEt})_2$,^{3b} $\text{Mg}[\text{Al}(\text{OEt})_4]_2$,^{3b} PhOMgX ,^{11b} and $\text{Me}_3\text{C}_6\text{H}_2\text{OMgX}$.^{11b} However, in these reactions, significant amounts of esters of the type $\text{RCO}_2\text{-CH}_2\text{R}$ derived by a dismutation of aldehydes are usually produced as by-products, together with significant amounts of aldol-condensation products. In the present study, we do

Table 1. The effect of the metal alkoxide on the aldol-Tishchenko reaction of the 3-phenylpropanal (**1a**) at 0°C

Entry	Base	Mol equiv	Time (h)	4a + 4a' (%)
1	<i>i</i> -PrOMgCl	0.33	12	81
2	<i>i</i> -PrOMgCl	0.17	12	75
3	<i>t</i> -BuOMgCl	0.33	4	73
4	<i>t</i> -BuOMgCl	0.17	5	61
5	<i>i</i> -PrOZnEt	0.33	12	91
6	<i>i</i> -PrOZnEt	0.17	12	73

not observe the above-mentioned side reactions when *i*-PrOZnEt was used. Therefore, we consider adapting the *i*-PrOZnEt as promoter to the aldol-Tishchenko reaction.

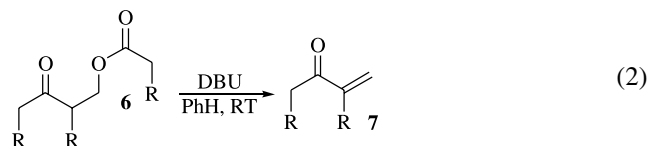
In order to evaluate the possibility of the further synthetic applications, the ratio of the aldol-Tishchenko products **4a** and **4a'** is an important factor. Unfortunately, due to the difficulty in separation and in analyzing the ¹H NMR spectrum, we did not know the ratio of products **4a** and **4a'** in each cases of the Table 1. Therefore, we tried to use an indirect method to estimate their ratio based on the yields of their oxidized products. A mixture of the aldol-Tishchenko products obtained from entry 5 (Table 1) were treated with PCC in the presence of 4 Å molecular sieves to give α,β -unsaturated aldehyde **5a** in 8% yield and β -acyloxyketone **6a** in 84% yield. Compounds **5a** and **6a** were easily separated by simple silica gel column chromatography. The ¹H NMR spectrum of the minor compound **5a** is identical to the one reported in the literature.¹⁵ Presumably, the primary alcohol **4a'** was oxidized by PCC to give the corresponding β -acyloxyaldehyde, which then simultaneously underwent β -elimination to give the conjugated aldehyde **5a** (Eq. 1). Therefore, we can estimate the mole ratio of **4a'** and **4a** approximately about 1:10.5. In other words, we have demonstrated that the *i*-PrOZnEt is an excellent reagent to promote the aldol-Tishchenko reaction from aliphatic aldehyde **1a** and the predominant product is 1,3-diol monoester **4a**, which could be converted to β -acyloxyketone **6a** in excellent yield.



2.2. Organozinc isopropoxide-promoted aldol-Tishchenko reaction and its application to the formation of the corresponding α -methylene ketone

In our laboratory, we are interested in developing the methodology to prepare the α -methylene-aldehydes,¹⁶

-ketones,¹⁷ -esters,¹⁸ -lactones¹⁹ and its application in the total synthesis of the methylenolactocin.²⁰ There are several different methods to prepare the α -methylene ketone. The most general and useful one was the acid-catalyzed Mannich reaction of the appropriate saturated ketone **9**, formaldehyde, and secondary amine hydrochloride,²¹ followed by thermal β -elimination from the Mannich base **8**²² or the corresponding quaternary ammonium derivatives (Pathway A, Fig. 3).²³ Previously, we reported an one pot process to carry out the α -methylenation of the aryl alkyl ketones with a mixture of dibromomethane and diethylamine under microwave irradiation. The mannich base was formed as a transient intermediate before elimination.¹⁷ Since, β -acyloxyketone **6** can be easily prepared from simple aldehyde **1** in two steps in high yield. We envisioned that the β -acyloxyketone **6** is a potential candidate to prepare α -methylene ketone **7** by the elimination reaction (Pathway B, Fig. 3). Furthermore, the eliminated by product, i.e., carboxylic acid **1'**, should be easily interconverted to the corresponding aldehyde **1**. We found that the β -acyloxyketone **6a** was treated with 1,8-diazabicyclo[5.4.0]undec-7-ene (DBU) in benzene at rt for 5 h to give the α -methylene ketone **7a** in 75% yield (R = -CH₂Ph, Eq. 2; entry 1, Table 2). Therefore, this proposed methodology to prepare the α -methylene ketone is completely different from the common approaches appeared in the literature.



i-PrOZnEt can also be applied to promote the aldol-Tishchenko reaction of the simple aliphatic aldehyde **1b** and aldehyde tethered with olefinic moieties (**1c** and **1d**) to give the corresponding 1,3-diol monoesters in good to excellent yields (entries 2–4). These 1,3-diol monoesters were oxidized by PCC followed by elimination with DBU to give the corresponding α -methylene ketones **7b–7d** in good yields (entries 2–4). In order to understand how general and versatile of our methodology in the aldol-Tishchenko reaction and in their further transformation to the α -methylene ketones, the starting aldehydes tethered with several different labile functionalities were employed. By similar reaction sequences, we were able to prepare the α -methylene ketones **7e** and **7f** tethered with methoxycarbonyl group (entries 5 and 6), **7g** and **7h** tethered with dimethylacetal group (entries 7 and 8), and **7i** tethered with

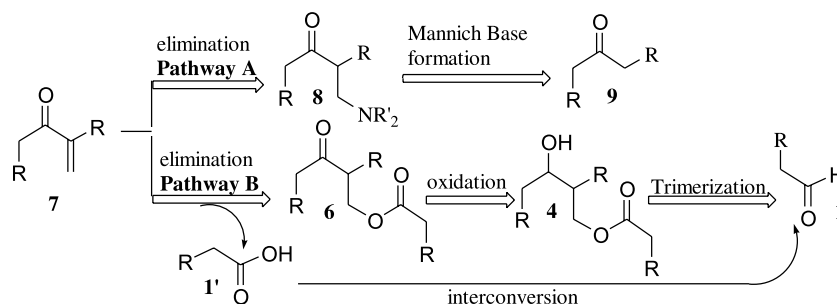
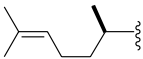
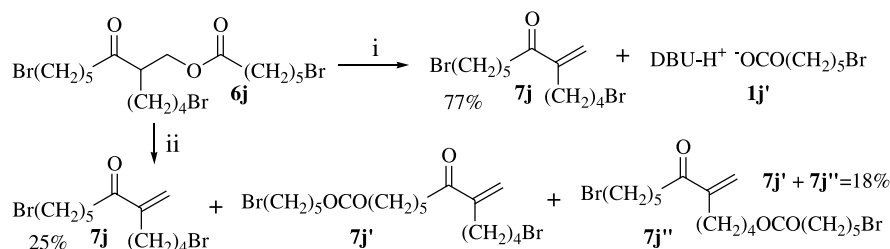


Figure 3. Retrosynthetic analysis of the preparation of the α -methylene ketone.

Table 2. *i*-PrOZnEt-promoted aldol-Tishchenko reaction of aldehydes, followed by oxidation with PCC and elimination with DBU

Entry	RCH ₂ CHO 1 R=	aldol-Tishchenko		Oxidation		Elimination	
		Time (h)	Yields (%)	Time (h)	Yield (%)	Time (h)	Yield (%)
1	PhCH ₂ – 1a	18	4a' + 4a 91	6	6a 84	5	7a 75
2	<i>n</i> -Pentyl– 1b	12	4b' + 4b 90	3	6b 75	6	7b 76
3	 1c	14	4c' + 4c 56	6	6c 80	7	7c 70
4	H ₂ C=CH(CH ₂) ₇ – 1d	18	4d' + 4d 84	5	6d 80	10	7d 71
5	MeO ₂ C(CH ₂) ₅ – 1e	24	4e' + 4e 68	8	6e 91	10	7e 75
6	MeO ₂ C(CH ₂) ₃ – 1f	24	4f' + 4f 50	12	6f 56	8	7f 80
7	(MeO) ₂ CH(CH ₂) ₅ – 1g	20	4g' + 4g 73	4	6g 64	7	7g 90
8	(MeO) ₂ CH(CH ₂) ₃ – 1h	22	4h' + 4h 72	4	6h 62	10	7h 92
9	MeCO(CH ₂) ₇ – 1i	15	4i' + 4i 50	8	6i 85	10	7i 80
10	Br(CH ₂) ₄ – 1j	20	4j' + 4j 87	12	6j 70	12	7j 77

**Scheme 1.** Reagents and conditions: (i) DBU, PhH, 0 °C to rt, 12 h; (ii) DBU, PhH, rt, 6 h.

acetyl group (entry 9) and **7j** tethered with bromo group (entry 10).

It is worthy to mention that the reaction temperature is crucial to the formation of the desired elimination product **7j** from compound **6j**. When the keto-ester **6j**, which was derived from the aldol-Tishchenko product of the 6-bromohexanal (**1j**), was treated with DBU at rt, the desired product **7j** was isolated in only 25% yield. We could also isolate an inseparable mixtures of ester–enone **7j'** and **7j''** in 18% yield (Scheme 1). The ratio of these two side products is 0.3:1 as estimated from their ¹H NMR chemical shifts at δ 4.14 and 4.06 ppm. These two characteristic peaks are the absorption of the methylene groups adjacent to the carboxyl group (–CH₂OCOR) of compounds **7j'** and **7j''**. However, we do not know the exact structure of the major product. The formations of **7j'** and **7j''** may be rationalized as follows. Each one of the bromides in compound **7j** is replaced by 6-bromocarboxylate **1j'**, a product formed from the elimination of compound **6j** by DBU, at rt. Fortunately, these side products can be avoided by carrying out the elimination reaction at 0 °C and the α-methylene ketone **7j** was formed in 77% yield (entry 10).

3. Conclusions

We have demonstrated that *i*-PrOZnEt is an excellent reagent to promote the aldol-Tishchenko reaction of aliphatic aldehydes tethered with several functional groups. Both 1,3-diol monoesters **4** and **4'** were formed and the primary alkyl carboxylate **4** was formed as the major product. The ratio of **4** and **4'** in each case are approximately 10:1. The major isomer **4** can be used for the preparation of the α-methylene ketones **7** by treatment with DBU.

Therefore, we have developed a new methodology to prepare the α-methylene ketones **7** from the simple aldehyde **1** via aldol-Tishchenko, oxidation and elimination sequences in good yields in three steps.

4. Experimental

All reactions were carried out under nitrogen. Unless otherwise noted, materials were obtained from commercial suppliers and used without further purification. The ¹H and ¹³C NMR spectra were recorded on a Bruker Avance DPX400 spectrometer, and chemical shifts were given in ppm downfield from tetramethylsilane (TMS). The ¹H and ¹³C NMR spectra were recorded on a Bruker Avance DPX-400 and chemical shifts were given in ppm downfield from tetramethylsilane (TMS). IR spectra were taken with a Perkin–Elmer 682 spectrophotometer and only noteworthy absorption was listed. Mass spectra were measured on a VG-Trio-2000GC/MS spectrometer (National Chiao-Tung University) by electronic impact at 70 eV (unless otherwise indicated). High Resolution Mass Spectroscopy (HRMS) was measured on a Finnigan/Thermo Quest MAT 95XL (National Chung Hsing University) and FAB Mass spectra were recorded with 3-nitrobenzyl alcohol matrix using argon or xenon as the target gas. Aldehydes used in this study were either commercially available or prepared by the literature method. The undec-10-enal (**1d**),²⁴ 10-oxoundecanal (**1i**),²⁵ and 6-bromohexanal (**1j**)^{16b} were prepared from their corresponding alcohols by PCC oxidation. 8-Oxooctanoic acid methyl ester (**1e**), 6-oxohexanoic acid methyl ester (**1f**), 8,8-dimethoxyoctanal (**1g**) and 6,6-dimethoxyhexanal (**1h**) were prepared according to the literature procedure from their corresponding cycloalkenes.²⁶

4.1. General procedure to carry out the aldol-Tishchenko reaction of the aldehyde and the mixture of the reaction products were subsequently oxidized by PCC

i-PrOZnEt was obtained from the reaction of Et₂Zn (1.0 M in hexane, 0.90 mL, 0.90 mmol) and isopropanol (0.07 mL, 0.90 mmol) in THF (3 mL) at rt for 5 min. To a solution of hydrocinnamaldehyde (**1a**) (603.8 mg, 4.5 mmol) in 10 mL of THF, *i*-PrOZnEt was added dropwise at 0 °C in a period of 5 min. The reaction was warmed slowly to rt and stirred at this temperature for 12 h. The reaction is quenched with 1 N HCl and extracted with ethyl ether. The organic layer was dried over MgSO₄, concentrated, and chromatographed on silica gel column to give a mixture of the 1,3-diol monoesters **4a'** and **4a** (549.5 mg, 4.09 mmol) in 91% yield. To a mixture of the 1,3-diol monoesters **4a'** and **4a** (549.5 mg, 4.09 mmol) in 10 mL of CH₂Cl₂, PCC (pyridinium chlorochromate, 1102 mg, 5.11 mmol) and anhydrous 4 Å molecular sieves (500 mg) was added at 0 °C. The reaction was warmed slowly to rt and stirred at this temperature for 3 h. To the crude reaction mixture, ether was added and most of the chromium salts were precipitated out. After filtration, the filtrate was concentrated, and chromatographed on silica gel column to give the β-acyloxyketone **6a** (1372 mg, 3.43 mmol) in 84% yield as a colorless oil and α,β-unsaturated aldehyde **5a**¹⁵ (82.5 mg, 0.33 mmol) in 8% yield.

4.1.1. 3-Phenylpropionic acid 2-benzyl-3-oxo-5-phenylpentyl ester (6a). TLC *R*_f=0.60 (hexane/EtOAc=10:1); ¹H NMR (CDCl₃, 400 MHz) δ 2.41–2.49 (m, 1H), 2.54–2.59 (m, 2H), 2.61–2.85 (m, 5H), 2.85–2.93 (m, 2H), 3.04–3.12 (m, 1H), 4.20 (d, *J*=6.5 Hz, 2H, –CH₂O₂C–), 7.06–7.29 (m, 15H, Ph-H); ¹³C NMR (CDCl₃, 100 MHz) δ 29.1 (2°), 30.8 (2°), 34.8 (2°), 35.6 (2°), 45.4 (2°), 52.6 (3°), 64.3 (2°, –CH₂O₂C–), 126.0 (3°), 126.3 (3°), 126.7 (3°), 128.2 (3°), 128.3 (3°), 128.4 (3°), 128.5 (3°), 128.7 (3°), 128.8 (3°), 138.1 (4°), 140.2 (4°), 140.9 (4°), 172.4 (4°, –CO₂–), 210.2 (4°, –C=O); IR (KBr, neat): 3061, 3027, 2928, 1737, 1720, 1604, 1496, 1454, 1371, 1245, 1160, 1078, 1030, 749 cm^{–1}; MS *m/z* (relative intensity): 400 (M⁺, 1), 250 (100), 91 (31); HRMS Calcd for C₂₇H₂₈O₃: 400.2038. Found: 400.2047.

4.1.2. Heptanoic acid 3-oxo-2-pentylonyl ester (6b). Followed the general procedure to prepare the aldol-Tishchenko products (**4b'** and **4b**) from the aldehyde **1b** in 90% yield. According to the general procedure, the aldol-Tishchenko products (**4b'** and **4b**) were then oxidized by PCC to give compound **6b** in 75% yield as a colorless oil; TLC *R*_f=0.72 (hexane/EtOAc=10:1); ¹H NMR (CDCl₃, 400 MHz) δ 0.83–0.87 (m, 9H), 1.21–1.40 (m, 18H), 1.53–1.59 (m, 6H), 2.23 (t, *J*=7.6 Hz, 2H, –CO₂CH₂–), 2.42 (t, *J*=7.2 Hz, 2H, –CH₂–), 2.80 (qu, *J*=6.8 Hz, 1H, –CHCO), 4.14 (d, *J*=6.8 Hz, 2H, AB system, –CH₂O₂C–); ¹³C NMR (CDCl₃, 100 MHz) δ 13.88 (1°), 13.93 (1°), 22.3 (2°), 22.4 (2°), 22.5 (2°), 23.3 (2°), 24.8 (2°), 26.7 (2°), 28.6 (2°), 28.7 (2°), 28.8 (2°), 31.4 (2°), 31.6 (2°), 31.8 (2°), 34.2 (2°), 43.0 (2°), 50.8 (3°), 64.5 (2°, –CH₂O₂C–), 173.4 (4°, C=O), 211.9 (4°, C=O); IR (KBr, neat): 2961, 2930, 2862, 1741 (C=O), 1720 (C=O), 1469, 1378, 1231, 1167, 1103 cm^{–1}; MS *m/z* (relative intensity): 340 (M⁺, 1), 113 (100), 85 (31); HRMS Calcd for C₂₁H₄₀O₃: 340.2977. Found: 340.2969.

4.1.3. 3,7-Dimethyloct-6-enoic acid 2-(1,5-dimethylhex-4-enyl)-5,9-dimethyl-3-oxodec-8-enyl ester (6c). Followed the general procedure to prepare the aldol-Tishchenko products (**4c'** and **4c**) from the aldehyde **1c** in 56% yield. According to the general procedure, the aldol-Tishchenko products (**4c'** and **4c**) were then oxidized by PCC to give compound **6c** in 80% yield as a colorless oil; TLC *R*_f=0.78 (hexane/EtOAc=10:1); ¹H NMR (CDCl₃, 400 MHz) δ 0.85–0.91 (m, 9H, –CH₃), 1.15–1.34 (m, 6H, –CH₂–), 1.58 (s, 6H, –CH₃), 1.60 (s, 3H, –CH₃), 1.67 (s, 6H, CH₃), 1.68 (s, 3H, CH₃), 1.88–2.07 (m, 10H), 2.21–2.27 (m, 2H, –CH₂–), 2.39 (dd, *J*=16.7, 5.1 Hz, 1H), 2.72–2.76 (m, 1H, –CH), 4.21–4.26 (m, 2H, –CH₂O₂C–), 5.04–5.09 (m, 3H, –CH=C(CH₃)₂); ¹³C NMR (CDCl₃, 100 MHz) δ 15.9 (1°), 17.6 (1°), 17.6 (1°), 17.7 (1°), 19.5 (1°), 19.7 (1°), 25.4 (2°), 25.5 (2°), 25.6 (2°), 25.7 (1°), 28.3 (1°), 29.93 (1°), 29.98 (3°), 32.2 (3°), 35.1 (2°), 36.8 (2°), 37.0 (2°), 41.7 (2°), 50.6 (2°), 55.0 (3°), 62.0 (2°, –CH₂O₂C–), 123.8 (3°), 124.2 (3°), 124.3 (3°), 131.4 (4°, –C=C(CH₃)₂), 131.5 (4°, –C=C(CH₃)₂), 131.9 (4°, –C=C(CH₃)₂), 173.0 (4°, –CO₂), 210.8 (4°, –C=O); IR (KBr, neat): 2967, 2925, 2851, 1739, 1720, 1457, 1378, 1286, 1244, 1152, 986 cm^{–1}; MS *m/z* (relative intensity): 461 (M⁺ + 1, 3), 272 (21), 290 (21), 109 (65); HRMS Calcd for C₃₀H₅₂O₃: 460.3916. Found: 460.3923.

4.1.4. Undec-10-enoic acid 2-dec-9-enyl-3-oxotridec-12-enyl ester (6d). Followed the general procedure to prepare the aldol-Tishchenko products (**4d'** and **4d**) from the aldehyde **1d** in 84% yield. According to the general procedure, the aldol-Tishchenko products (**4d'** and **4d**) were then oxidized by PCC to give compound **6d** in 80% yield as a colorless oil; TLC *R*_f=0.63 (hexane/EtOAc=10:1); ¹H NMR (CDCl₃, 400 MHz) δ 1.24–1.36 (m, 30H), 1.56–1.58 (m, 6H), 2.00 (pseudo q, *J*=6.6 Hz, 6H, CH₂=CHCH₂–), 2.22 (t, *J*=6.8 Hz, 2H, –CH), 2.41 (t, *J*=7.4 Hz, 2H), 2.79 (qu, *J*=6.9 Hz, 1H), 4.13 (d, *J*=6.8 Hz, 2H, –CH₂O₂C–), 4.88–4.97 (m, 6H, –CH=CH₂), 5.71–5.81 (m, 3H, –CH=CH₂); ¹³C NMR (CDCl₃, 100 MHz) δ 23.2 (2°), 24.8 (2°), 27.0 (2°), 28.5 (2°), 28.75 (2°), 28.80 (2°), 28.88 (2°), 28.96 (2°), 28.98 (2°), 29.00 (2°), 29.09 (2°), 29.10 (2°), 29.19 (2°), 29.23 (2°), 29.3 (2°), 29.5 (2°), 33.6 (2°), 33.7 (2°), 34.1 (2°), 42.9 (2°), 50.7 (3°), 64.5 (2°, –CH₂O₂C–), 114.0 (2°, –CH=CH₂), 114.1 (2°, –CH=CH₂), 138.9 (3°, –CH=CH₂), 139.0 (3°, –CH=CH₂), 173.3 (4°, –CO₂–), 211.7 (4°, –C=O); IR (KBr, neat): 3076, 2926, 2855, 1740, 1717, 1640, 1464, 1167, 994, 909 cm^{–1}; MS (FAB) *m/z* (relative intensity): 503 (M⁺ + 1, 5), 319 (55), 149 (25); HRMS (FAB) Calcd for [M + H]⁺ C₃₃H₅₉O₃: 503.4464. Found: 503.4453.

4.1.5. 7-(7-Methoxycarbonylheptanoyloxymethyl)-8-oxopentadecanedioic acid dimethyl ester (6e). Followed the general procedure to prepare the aldol-Tishchenko products (**4e'** and **4e**) from the aldehyde **1e** in 68% yield. According to the general procedure, the aldol-Tishchenko products (**4e'** and **4e**) were then oxidized by PCC to give compound **6e** in 91% yield as a colorless oil; TLC *R*_f=0.79 (hexane/EtOAc=1:1); ¹H NMR (CDCl₃, 400 MHz) δ 1.22–1.54 (m, 24H), 2.15–2.23 (m, 8H, –CH₂–), 2.37 (t, *J*=7.2 Hz, 2H), 2.73 (qu, *J*=6.8 Hz, 1H, –CH), 3.57 (s, 9H, –OCH₃), 4.07 (d, *J*=6.8 Hz, 2H, –CH₂O₂C–); ¹³C NMR (CDCl₃, 100 MHz) δ 22.8 (2°), 24.3 (2°), 24.4 (2°), 24.45 (2°), 24.47

(2°), 26.5 (2°), 28.1 (2°), 28.43 (2°), 28.45 (2°), 28.53 (2°), 28.7 (2°), 28.8 (2°), 33.57 (2°), 33.65 (2°), 33.67 (2°), 33.8 (2°), 42.7 (2°), 50.5 (3°), 51.1 (1°), 51.2 (1°), 64.2 (2°, $-\text{CH}_2\text{O}_2\text{C}-$), 173.0 (4°, $-\text{C}=\text{O}$), 173.65 (4°, $-\text{CO}_2$), 173.74 (4°, $-\text{CO}_2$), 173.75 (4°, $-\text{CO}_2$), 211.2 (4°, $-\text{C}=\text{O}$); IR (KBr, neat): 2938, 2860, 1739, 1437, 1362, 1172, 1011, 879, 731 cm^{-1} ; MS (FAB) m/z (relative intensity): 515 ($\text{M}^+ + 1$, 4), 307 (27), 154 (100), 137 (57); HRMS (FAB) Calcd for $[\text{M} + \text{H}]^+ \text{C}_{27}\text{H}_{47}\text{O}_9$: 515.3220. Found: 515.3228.

4.1.6. 5-(5-Methoxycarbonylpentanoyloxymethyl)-6-oxoundecanedioic acid dimethyl ester (6f). Followed the general procedure to prepare the aldol-Tishchenko products (**4f'** and **4f**) from the aldehyde **1f** in 50% yield. According to the general procedure, the aldol-Tishchenko products (**4f'** and **4f**) were then oxidized by PCC to give compound **6f** in 56% yield as a colorless oil; TLC R_f =0.50 (ether/hexane=2:1); ^1H NMR (CDCl_3 , 400 MHz) δ 1.56–1.64 (m, 12H), 2.27–2.34 (m, 8H, $-\text{CH}_2$), 2.48–2.51 (m, 2H, $-\text{CH}_2$), 2.82 (qu, J =6.4 Hz, 1H, $-\text{CH}$), 3.67 (s, 9H, $-\text{OCH}_3$), 4.18 (d, J =6.3 Hz, 2H, $-\text{CH}_2\text{O}_2\text{C}-$); ^{13}C NMR (CDCl_3 , 100 MHz) δ 22.3 (2°), 22.6 (2°), 24.16 (2°), 24.20 (2°), 24.3 (2°), 27.8 (2°), 33.50 (2°), 33.58 (2°), 33.67 (2°), 33.70 (2°), 42.3 (2°), 50.5 (3°), 51.4 (1°, $-\text{OCH}_3$), 51.5 (1°, $-\text{OCH}_3$), 64.2 (2°, $-\text{CH}_2\text{O}_2\text{C}-$), 172.8 (4°, $-\text{CO}_2\text{CH}_3$), 173.3 (4°, $-\text{CO}_2\text{CH}_3$), 173.55 (4°, $-\text{CO}_2\text{CH}_3$), 173.64 (4°, $-\text{CO}_2\text{CH}_3$), 210.6 (4°, $-\text{C}=\text{O}$); IR (KBr, neat): 2952, 1738, 1436, 1368, 1171 cm^{-1} ; MS (FAB) m/z (relative intensity): 431 ($\text{M}^+ + 1$, 29), 271 (55), 239 (38), 154 (100), 137 (81), 111 (46); HRMS (FAB) Calcd for $\text{C}_{21}\text{H}_{35}\text{O}_9$ $[\text{M} + \text{H}]^+$: 431.2281. Found: 431.2272.

4.1.7. 8,8-Dimethoxyoctanoic acid 2-(6,6-dimethoxyheptyl)-10,10-dimethoxy-3-oxodecyl ester (6g). Followed the general procedure to prepare the aldol-Tishchenko products (**4g'** and **4g**) from the aldehyde **1g** in 73% yield. According to the general procedure, the aldol-Tishchenko products (**4g'** and **4g**) were then oxidized by PCC to give compound **6g** in 64% yield as a pale yellow oil; TLC R_f =0.83 (ether/hexane=3:1); ^1H NMR (CDCl_3 , 400 MHz) δ 1.20–1.35 (m, 20H), 1.45–1.55 (m, 10H), 2.20 (t, J =7.5 Hz, 2H, $-\text{OC}=\text{OCH}_2-$), 2.39 (t, J =7.3 Hz, 2H, $-\text{CH}_2\text{CO}-$), 2.76 (qu, J =6.8 Hz, 1H, $-\text{CHC}=\text{O}$), 3.25 (s, 18H, $-\text{OCH}_3$), 4.10 (d, J =6.8 Hz, 2H, $-\text{CH}_2\text{O}_2\text{C}-$), 4.27–4.31 (m, 3H, $-\text{CH}(\text{OCH}_3)_2$); ^{13}C NMR (CDCl_3 , 100 MHz) δ 23.1 (2°), 24.2 (2°), 24.28 (2°), 24.31 (2°), 24.6 (2°), 27.0 (2°), 28.4 (2°), 28.89 (2°), 28.96 (2°), 29.01 (2°), 29.2 (2°), 29.4 (2°), 32.27 (2°), 32.31 (2°), 32.33 (2°), 34.0 (2°), 42.9 (2°), 50.7 (3°), 52.49 (1°, $-\text{OCH}_3$), 52.52 (1°, $-\text{OCH}_3$), 64.4 (2°, $-\text{CH}_2\text{O}_2\text{C}-$), 104.3 (3°), 104.4 (3°), 173.3 (4°, $-\text{CO}_2$), 211.6 (4°, $-\text{C}=\text{O}$); IR (KBr, neat): 2939, 1739, 1718 ($\text{C}=\text{O}$), 1463, 1385, 1127, 728 cm^{-1} ; MS m/z (relative intensity): 561 ($\text{M}^+ - 1$, 3), 435 (36), 403 ($\text{M}^+ - 159$), 391 (16), 307 (18), 154 (100); HRMS Calcd for ($\text{M}^+ - 159$) $\text{C}_{21}\text{H}_{39}\text{O}_7$: 403.2696. Found: 403.2704.

4.1.8. 6,6-Dimethoxyhexanoic acid 2-(4,4-dimethoxybutyl)-8,8-dimethoxy-3-oxooctyl ester (6h). Followed the general procedure to prepare the aldol-Tishchenko products (**4h'** and **4h**) from the aldehyde **1h** in 72% yield. According to the general procedure, the aldol-Tishchenko products (**4h'** and **4h**) were then oxidized by PCC to give compound **6h** in 62% yield as a pale yellow oil; TLC R_f =0.32 (ether/hexane=2:1); ^1H NMR (CDCl_3 , 400 MHz) δ 1.26–1.33 (m,

6H), 1.49–1.57 (m, 12H), 2.21 (t, J =7.5 Hz, 2H, $-\text{CH}_2$), 2.41 (t, J =7.2 Hz, 2H, $-\text{CH}_2$), 2.76 (qu, J =6.7 Hz, 1H, $-\text{CH}$), 3.24 (s, 18H, $-\text{OCH}_3$), 4.10 (d, J =7.6 Hz, 2H, $-\text{CH}_2\text{O}_2\text{C}-$), 4.25–4.30 (m, 3H, $-\text{CH}(\text{OCH}_3)_2$); ^{13}C NMR (CDCl_3 , 100 MHz) δ 22.1 (2°), 23.0 (2°), 23.97 (2°), 24.02 (2°), 24.5 (2°), 28.2 (2°), 32.0 (2°), 32.2 (2°), 32.4 (2°), 33.9 (2°), 42.8 (2°), 50.6 (3°), 52.58 (1°, $-\text{OCH}_3$), 52.64 (1°, $-\text{OCH}_3$), 52.8 (1°, $-\text{OCH}_3$), 64.3 (2°, $-\text{CH}_2\text{O}_2\text{C}-$), 104.0 (3°), 104.15 (3°), 104.19 (3°), 173.0 (4°, $-\text{CO}_2$), 211.2 (4°, $-\text{C}=\text{O}$); IR (KBr, neat): 2947, 2830, 1739, 1462, 1386, 1197, 1127, 1054, 957, 736 cm^{-1} ; MS (FAB) m/z (relative intensity): 478 ($\text{M}^+ + 1$, 2), 351 (28), 127 (73), 75 (100); HRMS (FAB) Calcd for $\text{C}_{24}\text{H}_{46}\text{O}_9$: 478.3142. Found: 478.3132.

4.1.9. 10-Oxoundecanoic acid 3,12-dioxo-2-(8-oxononyl)-tridecyl ester (6i). Followed the general procedure to prepare the aldol-Tishchenko products (**4i'** and **4i**) from the aldehyde **1i** in 50% yield. According to the general procedure, the aldol-Tishchenko products (**4i'** and **4i**) were then oxidized by PCC to give compound **6i** in 85% yield as a white solid; mp 55 °C; TLC R_f =0.47 (hexane/EtOAc=3:1); ^1H NMR (CDCl_3 , 400 MHz) δ 1.15–1.25 (m, 24H), 1.45–1.48 (m, 12H), 2.03 (s, 9H, $-\text{CH}_3$), 2.16 (t, J =7.5 Hz, 2H, $-\text{O}=\text{CCH}_2-$), 2.30–2.37 (m, 8H, $-\text{CH}_2$), 2.72 (qu, J =6.8 Hz, 1H, $-\text{CH}$), 4.06 (d, J =6.8 Hz, 2H, $-\text{CH}_2\text{O}_2\text{C}-$); ^{13}C NMR (CDCl_3 , 100 MHz) δ 23.1 (2°), 23.5 (2°), 23.6 (1°), 24.6 (2°), 26.8 (2°), 28.3 (2°), 28.80 (2°), 28.83 (2°), 28.86 (2°), 28.88 (2°), 28.90 (2°), 28.96 (2°), 29.00 (2°), 29.03 (2°), 29.2 (2°), 29.6 (2°), 33.9 (2°), 42.7 (2°), 43.4 (2°), 43.5 (2°), 50.6 (3°), 64.3 (2°, $-\text{CH}_2\text{O}_2\text{C}-$), 173.2 (4°, $-\text{CH}_2\text{CO}_2-$), 208.79 (4°, $-\text{C}=\text{O}$), 208.84 (4°, $-\text{C}=\text{O}$), 211.6 (4°, $-\text{C}=\text{O}$); IR (KBr, neat): 2931, 2856, 1730, 1717, 1460, 1413, 1361, 1233, 1168, 735 cm^{-1} ; MS (FAB) m/z (relative intensity): 551 ($\text{M}^+ + 1$, 13), 351 (100), 183 (51), 137 (25); HRMS (FAB) Calcd for $[\text{M} + \text{H}]^+ \text{C}_{33}\text{H}_{59}\text{O}_6$: 551.4312. Found: 551.4321.

4.1.10. 6-Bromohexanoic acid 8-bromo-2-(4-bromobutyl)-3-oxooctyl ester (6j). Followed the general procedure to prepare the aldol-Tishchenko products (**4j'** and **4j**) from the aldehyde **1j** in 87% yield. According to the general procedure, the aldol-Tishchenko products (**4j'** and **4j**) were then oxidized by PCC to give compound **6j** in 70% yield as a pale yellow oil; TLC R_f =0.51 (hexane/EtOAc=5:1); ^1H NMR (CDCl_3 , 400 MHz) δ 1.39–1.45 (m, 7H), 1.57–1.62 (m, 5H), 1.81–1.86 (m, 6H), 2.27 (t, J =7.4 Hz, 2H, $-\text{O}=\text{CCH}_2-$), 2.47 (td, J =7.2, 1.8 Hz, 2H), 2.80 (quin, J =5.6 Hz, 1H), 3.37–3.40 (m, 6H, $-\text{CH}_2\text{Br}$), 4.15 (t, J =5.2 Hz, 2H, $-\text{CH}_2\text{O}_2\text{C}-$); ^{13}C NMR (CDCl_3 , 100 MHz) δ 22.3 (2°), 23.9 (2°), 25.6 (2°), 27.51 (2°), 27.53 (2°), 27.6 (2°), 32.3 (2°), 32.5 (2°), 33.1 (2°), 33.4 (2°), 33.5 (2°), 33.8 (2°), 42.6 (2°), 50.7 (3°), 64.3 (2°, $\text{CH}_2\text{O}_2\text{C}-$), 172.9 (4°, $-\text{C}=\text{O}$), 210.9 (4°, $-\text{C}=\text{O}$); IR (KBr, neat): 2938, 2863, 1736, 1720, 1637, 1457, 1366, 1266, 1173, 737 cm^{-1} ; MS (FAB) m/z (relative intensity): 539 (5), 537 (10), 535 (11), 533 (5), 453 (8), 341 (37), 177 (49), 137 (18); HRMS (FAB) Calcd $[\text{M} + \text{H}]^+$ for $\text{C}_{18}\text{H}_{32}\text{O}_3^{79}\text{Br}_3$: 532.9902. Found: 532.9910.

4.2. General procedure to prepare the α -methylene ketone via elimination of the β -acyloxyketone

To a solution of the β -acyloxyketone **6a** (152 mg,

0.38 mmol) in 5 mL of benzene was added DBU (0.06 mL, 0.42 mmol) and the reaction mixture was stirred at rt for 5 h. The reaction mixture was concentrated to give the crude residues, which were chromatographed on silica gel column to give the α -methylene ketone **7a** (71.3 mg, 0.29 mmol) in 75% yield as a pale yellow oil.

4.2.1. 2-Benzyl-5-phenylpent-1-en-3-one (7a).^{22d} TLC R_f =0.66 (hexane/EtOAc=10:1); ^1H NMR (CDCl_3 , 400 MHz) δ 2.99–3.07 (m, 4H, $-\text{CH}_2$), 3.69 (s, 2H), 5.67 (t, J =1.32 Hz, 1H, $-\text{C}=\text{CH}_2$), 6.10 (s, 1H, $-\text{C}=\text{CH}_2$), 7.22–7.35 (m, 10H, Ph-H); ^{13}C NMR (CDCl_3 , 100 MHz) δ 30.2 (2°), 37.0 (2°), 39.5 (2°), 125.2 (3°), 125.95 (3°), 126.13 (2°), $-\text{C}=\text{CH}_2$, 128.25 (3°), 128.33 (3°), 128.4 (3°), 129.0 (3°), 139.0 (4°), 141.1 (4°), 148.2 (4°, $-\text{C}=\text{CH}_2$), 200.1 (4°, $-\text{C}=\text{O}$); IR (KBr, neat): 3085, 3061, 3027, 2925, 1678, 1627, 1603, 1495, 1453, 1432, 1409, 1368, 1078, 941, 745, 700 cm^{-1} ; MS m/z (relative intensity): 250 (M^+ , 100), 91 (22); HRMS Calcd for $\text{C}_{18}\text{H}_{18}\text{O}$: 250.1358. Found: 250.1349.

4.2.2. 2-Pentyl-non-1-en-3-one (7b). Followed the general procedure to carry out the elimination of β -acyloxy ketone **6b** to give α -methylene ketone **7b** in 76% yield as a pale yellow oil; TLC R_f =0.70 (hexane/EtOAc=10:1); ^1H NMR (CDCl_3 , 400 MHz) δ 0.85–0.88 (m, 6H), 1.24–1.40 (m, 12H), 1.53–1.59 (m, 2H), 2.24 (t, J =7.8 Hz, 2H, $-\text{CH}_2\text{C}=\text{CH}_2$), 2.64 (t, J =7.5 Hz, 2H), 5.67 (s, 1H, $-\text{C}=\text{CH}_2$), 5.94 (s, 1H, $-\text{C}=\text{CH}_2$); ^{13}C NMR (CDCl_3 , 100 MHz) δ 13.9 (1°), 14.0 (1°), 22.4 (2°), 22.5 (2°), 24.6 (2°), 28.1 (2°), 29.0 (2°), 30.9 (2°), 31.5 (2°), 31.6 (2°), 37.8 (2°), 123.1 (2°, $-\text{C}=\text{CH}_2$), 149.2 (2°, $-\text{C}=\text{CH}_2$), 202.4 (4°, $-\text{C}=\text{O}$); IR (KBr, neat): 3095, 2928, 2857, 1739, 1679 ($\text{C}=\text{O}$), 1623, 1464, 1378, 1121, 931 cm^{-1} ; MS m/z (relative intensity): 210 (M^+ , 100), 125 (11); HRMS Calcd for $\text{C}_{14}\text{H}_{26}\text{O}$: 210.1984. Found: 210.1976.

4.2.3. (6R,10R)-2,6,10,14-Tetramethyl-7-methylene-pentadeca-2,13-dien-8-one (7c). Followed the general procedure to carry out the elimination of β -acyloxy ketone **6c** to give α -methylene ketone **7c** in 70% yield as a pale yellow oil; TLC R_f =0.83 (hexane/EtOAc=20:1); ^1H NMR (CDCl_3 , 400 MHz) δ 0.89 (d, J =6.6 Hz, 3H, $-\text{CH}_3$), 1.01 (d, J =6.8 Hz, 3H, $-\text{CH}_3$), 1.19–1.48 (m, 5H), 1.56 (s, 3H, $\text{C}=\text{C}(\text{CH}_3)_2$), 1.59 (s, 3H, $\text{C}=\text{C}(\text{CH}_3)_2$), 1.67 (s, 6H, $\text{C}=\text{C}(\text{CH}_3)_2$), 1.87–2.07 (m, 5H), 2.46 (dd, J =8.1 Hz, 1H), 2.63–2.68 (m, 1H), 2.78–2.84 (m, 1H), 5.08 (br t, 2H, $=\text{CH}(\text{CH}_3)_2$), 5.66 (s, 1H, $=\text{CH}_2$), 5.98 (s, 1H, $-\text{C}=\text{CH}_2$); ^{13}C NMR (CDCl_3 , 100 MHz) δ 17.6 (1°), 19.8 (1°), 20.1 (1°), 20.2 (1°), 25.5 (2°), 25.7 (1°), 25.9 (2°), 29.7 (3°), 32.6 (3°), 36.3 (2°), 37.2 (2°), 45.5 (2°), 121.8 (2°), 124.40 (3°, $-\text{CH}=\text{C}(\text{CH}_3)_2$), 124.43 (3°, $-\text{CH}=\text{C}(\text{CH}_3)_2$), 131.40 (4°, $-\text{CH}=\text{C}(\text{CH}_3)_2$), 131.43 (4°, $-\text{CH}=\text{C}(\text{CH}_3)_2$), 154.8 (4°, $-\text{C}=\text{CH}_2$), 202.2 (4°, $-\text{C}=\text{O}$); $[\alpha]_D^{28}$ = -32.7 (c =3.0 $\times$ 10 $^{-4}$, CH_2Cl_2); IR (KBr, neat): 3095, 2967, 2925, 2851, 1677, 1457, 1378, 931 cm^{-1} ; MS m/z (relative intensity): 290 (M^+ , 13), 233 (20), 207 (31), 165 (38), 109 (70); HRMS Calcd for $\text{C}_{20}\text{H}_{34}\text{O}$: 290.2610. Found: 290.2613.

4.2.4. 12-Methylenedocosa-1,21-dien-11-one (7d). Followed the general procedure to carry out the elimination of β -acyloxy ketone **6d** to give α -methylene ketone **7d** in 71% yield as a pale yellow oil; TLC R_f =0.92 (hexane/

EtOAc=30:1); ^1H NMR (CDCl_3 , 400 MHz) δ 1.27–1.38 (m, 20H), 1.54–1.59 (m, 2H), 2.01 (pseudo q, J =6.8 Hz, 4H, $\text{CH}_2=\text{CHCH}_2$), 2.23 (t, J =7.5 Hz, 2H, $-\text{CH}_2\text{C}=\text{O}$), 2.63 (t, J =7.4 Hz, 2H, $-\text{CH}_2$), 4.88–4.98 (m, 4H, $-\text{CH}=\text{CH}_2$), 5.65 (s, 1H, $-\text{C}=\text{CH}_2$), 5.72–5.82 (m, 2H, $-\text{CH}=\text{CH}_2$), 5.92 (s, 1H, $-\text{CH}=\text{CH}_2$); ^{13}C NMR (CDCl_3 , 100 MHz) δ 24.6 (2°), 28.4 (2°), 28.8 (2°), 28.98 (2°), 29.00 (2°), 29.20 (2°), 29.25 (2°), 29.33 (2°), 30.9 (2°), 33.7 (2°), 37.8 (2°), 114.0 (2°, $-\text{CH}=\text{CH}_2$), 123.1 (4°, $-\text{C}=\text{CH}_2$), 139.0 (3°, $-\text{CH}=\text{CH}_2$), 149.1 (4°, $-\text{C}=\text{CH}_2$), 202.2 (4°, $-\text{C}=\text{O}$); IR (KBr, neat): 3078, 2927, 2855, 1679, 1643, 1469, 1433, 1366, 909 cm^{-1} ; MS (FAB) m/z (relative intensity): 319 ($\text{M}+1$, 46), 179 (9), 149 (13), 109 (22); HRMS (FAB) Calcd for $[\text{M}+\text{H}]^+$ $\text{C}_{22}\text{H}_{39}\text{O}$: 319.3001. Found: 319.3004.

4.2.5. 7-Methylene-8-oxo-pentadecanedioic acid dimethyl ester (7e). Followed the general procedure to carry out the elimination of β -acyloxy ketone **6e** to give α -methylene ketone **7e** in 75% yield as a pale yellow oil; TLC R_f =0.83 (hexane/EtOAc=10:1); ^1H NMR (CDCl_3 , 400 MHz) δ 1.27–1.60 (m, 14H), 2.20–2.27 (m, 6H, $-\text{CH}_2$), 2.61 (t, J =7.4 Hz, 2H, $-\text{CH}_2$), 3.61 (s, 6H, $-\text{OCH}_3$), 5.65 (s, 1H, $-\text{C}=\text{CH}_2$), 5.92 (s, 1H, $-\text{C}=\text{CH}_2$); ^{13}C NMR (CDCl_3 , 100 MHz) δ 24.3 (2°), 24.59 (2°), 24.64 (2°), 28.0 (2°), 28.7 (2°), 28.80 (2°), 28.81 (2°), 30.6 (2°), 33.8 (2°), 37.5 (2°), 51.3 (1°, $-\text{OCH}_3$), 123.5 (2°, $-\text{C}=\text{CH}_2$), 148.7 (4°, $-\text{C}=\text{CH}_2$), 174.00 (4°, $-\text{CO}_2$), 174.01 (4°, $-\text{CO}_2$), 201.9 (4°, $-\text{C}=\text{O}$); IR (KBr, neat): 2940, 2860, 1739, 1678, 1619, 1437, 1361, 1257, 1173, 1005, 938 cm^{-1} ; MS (FAB) m/z (relative intensity): 327 (M^++1 , 60), 149 (100), 137 (71), 107 (27); HRMS (FAB) Calcd for $[\text{M}+\text{H}]^+$ $\text{C}_{18}\text{H}_{31}\text{O}_5$: 327.2171. Found: 327.2180.

4.2.6. 5-Methylene-6-oxoundecanedioic acid dimethyl ester (7f). Followed the general procedure to carry out the elimination of β -acyloxy ketone **6f** to give α -methylene ketone **7f** in 80% yield as a pale yellow oil; TLC R_f =0.70 (ether/hexane=2:1); ^1H NMR (CDCl_3 , 400 MHz) δ 1.28–1.69 (m, 6H), 2.21–2.27 (m, 6H, $-\text{CH}_2$), 2.64 (br t, 2H, $-\text{CH}_2$), 3.59–3.60 (m, 6H, $-\text{OCH}_3$), 5.70 (s, 1H, $-\text{C}=\text{CH}_2$), 5.96 (s, 1H, $-\text{C}=\text{CH}_2$); ^{13}C NMR (CDCl_3 , 100 MHz) δ 23.5 (2°), 23.7 (2°), 24.4 (2°), 30.1 (2°), 33.4 (2°), 33.7 (2°), 37.1 (2°), 51.3 (1°, $-\text{OCH}_3$), 124.2 (2°, $-\text{C}=\text{CH}_2$), 147.8 (4°, $-\text{C}=\text{CH}_2$), 173.6 (4°, $-\text{CO}_2$), 173.7 (4°, $-\text{CO}_2$), 201.0 (4°, $-\text{C}=\text{O}$); IR (KBr, neat): 2952, 1737, 1677, 1437, 1368, 1199, 1173, 1005, 938 cm^{-1} ; MS m/z (relative intensity): 270 (M^+ , 7), 206 (21), 123 (89), 95 (100); HRMS Calcd for $\text{C}_{14}\text{H}_{22}\text{O}_5$: 270.1467. Found: 270.1476.

4.2.7. 2-(6,6-Dimethoxyhexyl)-10,10-dimethoxydec-1-en-3-one (7g). Followed the general procedure to carry out the elimination of β -acyloxy ketone **6g** to give α -methylene ketone **7g** in 90% yield as a pale yellow oil; TLC R_f =0.50 (hexane/EtOAc=3:1); ^1H NMR (CDCl_3 , 400 MHz) δ 1.30–1.39 (m, 12H), 1.53–1.57 (m, 6H), 2.23 (t, J =7.0 Hz, 2H, $-\text{CH}_2$), 2.63 (t, J =7.4 Hz, 2H, $-\text{CH}_2$), 3.28 (s, 12H, $-\text{OCH}_3$), 4.32 (t, J =5.7 Hz, 2H, $-\text{CH}(\text{OCH}_3)_2$), 5.66 (s, 1H, $-\text{CH}=\text{CH}_2$), 5.93 (s, 1H, $-\text{C}=\text{CH}_2$); ^{13}C NMR (CDCl_3 , 100 MHz) δ 24.3 (2°), 24.4 (2°), 24.5 (2°), 28.4 (2°), 29.13 (2°), 29.19 (2°), 29.23 (2°), 30.8 (2°), 32.39 (2°), 32.41 (2°), 37.7 (2°, $-\text{CH}_2\text{C}=\text{O}$), 52.6 (1°, $-\text{OCH}_3$), 104.5 (3°, $-\text{CH}(\text{OCH}_3)_2$), 123.3 (2°, $-\text{C}=\text{CH}_2$), 149.0 (4°,

–C=CH₂), 202.2 (4°, –C=O); IR (KBr, neat): 3060, 2938, 2861, 1739, 1676, 1467, 1394, 1370, 1273, 1127, 1048, 939, 739 cm^{–1}; MS *m/z* (relative intensity): 357 (M⁺ – 1, 3), 296 (M⁺ – 62, 6), 263 (60), 154 (20), 149 (36), 75 (100); HRMS Calcd for (M⁺ – 62) C₁₈H₃₂O₃: 296.2351. Found: 296.2347.

4.2.8. 2-(4,4-Dimethoxybutyl)-8,8-dimethoxyoct-1-en-3-one (7h). Followed the general procedure to carry out the elimination of β-acyloxy ketone **6h** to give α-methylene ketone **7h** in 92% yield a pale yellow oil; TLC *R*_f = 0.72 (ether/hexane = 4:1); ¹H NMR (CDCl₃, 400 MHz) δ 1.31–1.41 (m, 4H), 1.52–1.60 (m, 6H), 2.23 (t, *J* = 7.6 Hz, 2H, CH₂C=C), 2.63 (t, *J* = 7.3 Hz, 2H, –CH₂C=O), 3.26 (s, 12H, –OCH₃), 4.31 (t, *J* = 5.6 Hz, 2H, –CH(OCH₃)₂), 5.68 (s, 1H, –C=CH₂), 5.94 (s, 1H, –C=CH₂); ¹³C NMR (CDCl₃, 100 MHz) δ 23.3 (2°), 24.20 (2°), 24.22 (2°), 30.5 (2°), 32.1 (2°), 32.3 (2°), 37.5 (2°, –CH₂C=O), 52.61 (1°, –OCH₃), 52.64 (1°, –OCH₃), 104.3 (3°, –CH(OCH₃)₂), 123.7 (2°, –C=CH₂), 148.5 (4°, –C=CH₂), 201.7 (4°, –C=O); IR (KBr, neat): 2947, 2830, 1738, 1680, 1458, 1386, 1192, 1128, 1072, 953 cm^{–1}; MS (FAB) *m/z* (relative intensity): 302 (M⁺, 4), 271 (10), 147 (43), 75 (100); HRMS (FAB) Calcd for C₁₆H₃₀O₅: 302.2093. Found: 302.2095.

4.2.9. 10-Methyleneceneicosane-2,11,20-trione (7i). Followed the general procedure to carry out the elimination of β-acyloxy ketone **6i** to give α-methylene ketone **7i** in 80% yield as a white solid; mp 57–58 °C; TLC *R*_f = 0.53 (hexane/EtOAc = 3:1); ¹H NMR (CDCl₃, 400 MHz) δ 1.22–1.35 (m, 16H), 1.40–1.60 (m, 6H), 2.09 (s, 6H, –COCH₃), 2.20 (t, *J* = 7.5 Hz, 2H, –CH₂), 2.37 (t, *J* = 7.4 Hz, 4H, –CH₂), 2.61 (t, *J* = 7.4 Hz, 2H, –CH₂), 5.65 (s, 1H, –C=CH₂), 5.92 (s, 1H, –C=CH₂); ¹³C NMR (CDCl₃, 100 MHz) δ 23.7 (2°), 24.5 (2°), 28.3 (2°), 28.99 (2°), 29.01 (2°), 29.07 (2°), 29.13 (2°), 29.2 (2°), 29.7 (2°), 30.8 (2°), 37.7 (2°), 43.7 (2°), 123.3 (2°, –C=CH₂), 149.0 (4°, –C=CH₂), 202.3 (4°, –C=O), 209.1 (4°, –C=O); IR (KBr, neat): 3057, 2929, 2855, 1714, 1676, 1627, 1410, 1361, 1269, 1166, 937, 737 cm^{–1}; MS (FAB) *m/z* (relative intensity): 351 (M⁺ + 1, 12), 111 (14), 83 (45); HRMS (FAB) Calcd for [M + H]⁺ C₂₂H₃₉O₃: 351.2899. Found: 351.2894.

4.2.10. 10-Bromo-2-(4-bromobutyl)-oct-1-en-3-one (7j). Followed the general procedure to carry out the elimination of β-acyloxy ketone **6j** to give α-methylene ketone **7j** in 77% yield as a pale yellow oil; TLC *R*_f = 0.82 (hexane/EtOAc = 5:1); ¹H NMR (CDCl₃, 400 MHz) δ 1.43–1.66 (m, 6H, –CH₂), 1.84–1.90 (m, 4H, –CH₂C=O), 2.29 (t, *J* = 7.7 Hz, 2H, –CH₂–C=CH₂), 2.70 (t, *J* = 7.3 Hz, 2H, –CH₂), 3.40 (t, *J* = 6.8 Hz, 4H, –CH₂Br), 5.75 (s, 1H, –C=CH₂), 6.00 (s, 1H, –C=CH₂); ¹³C NMR (CDCl₃, 100 MHz) δ 23.6 (2°), 27.1 (2°), 27.8 (2°), 30.0 (2°), 32.4 (2°), 32.6 (2°), 33.4 (2°), 33.5 (2°), 37.4 (2°), 124.0 (2°, –C=CH₂), 148.3 (4°, –C=CH₂), 201.5 (4°, –C=O); IR (KBr, neat): 3102, 2937, 2861, 1677, 1626, 1437, 1366, 1265, 937, 738 cm^{–1}; MS (FAB) *m/z* (relative intensity): 343 (36), 341 (100), 339 (68), 261 (50), 259 (51); HRMS Calcd [M + H]⁺ for C₁₂H₂₀⁷⁹Br₂O: 338.9959. Found: 338.9952.

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